

Practical no 16

ALP to count number of '0s' and '1s' in a given number.

```
DATA SEGMENT
    N1 DW 5030H
DATA ENDS
CODE SEGMENT ASSUME
CS:CODE, DS:DATA START:
    MOV AX, DATA
    MOV DS, AX
    MOV AX, N1
    MOV CL, 16
BACK:
    SHL AX, 01H
    JC ONE
    INC BX
    JMP NXT ONE:
    INC DX
NXT
    DEC CX
    JNZ BACK
    INT 21H
CODE ENDS
END START
```

Practical no 17

Write an assembly language program to solve $p = x^2 + y^2$ using Macro.(x and y are 8 bit numbers.)

```
.MODEL SMALL
PROG MACRO a,b
MOV al,a
MUL al
MOV bl,al
MOV al,b
MUL al
ADD al,bl
ENDM
.DATA x DB 02H y DB 03H p DB DUP()
.CODE START:
MOV ax,data
MOV ds,ax
PROG x, y
MOV p,al
MOV ah,4Ch
Int 21H
END
```